Universität Münster

ERCIS

Bike count prediction as a time-series supervised learning task

Direct forecasts for +1h and +24h, with feature engineering based only on information available at forecast time t.

1

Direct targets

For horizon h, target = BikeCount.shift(-h). +24h predicts one hour 24 hours ahead, not a 24h sum.

2

Leakage-safe history

+1h uses lags 1, 2, 3, 24, 168. +24h starts at lag 24 because future counts after t are unknown.

3

Temporal validation

Last 61 days as holdout. Random splits are avoided because they leak time information through lags.

Feature set

Calendar	target hour, weekday, month, weekend
Cyclical	sin/cos for hour and month
Weather	7 buckets + temp, humidity, rain, wind
BikeCount	known lags and rolling means

Models and evaluation

- Ridge, RandomForest, XGBoost variants, MLP
- Rolling means: 3h / 24h for +1h; 24h for +24h
- One corrupt duplicate timestamp removed

8,759
clean rows

26
+1h features

61d
holdout

Submitted model and notebook evaluate +1h only; slides compare both requested horizons.

Boosted trees were strongest; XGBoost tuned log1p is our +1h submission model

Validation MSE on the last 61 days; lower is better. The table compares both required forecast horizons.

Model	+1h MSE	+24h MSE
naive horizon lag	22,138	60,513
naive lag168	45,525	45,525
Ridge	11,582	28,223
RandomForest	4,035	25,438
XGBoost	3,455	24,373
XGBoost log1p	3,320	22,595
XGBoost tuned	3,332	22,660
XGBoost tuned log1p	**3,220**	**19,988**
MLP	9,303	39,635

Selected model

XGBoost tuned log1p

+1h validation MSE: 3,220

Takeaways

All trained models beat the simple horizon-lag baseline. Boosted trees worked best on this small tabular time series.

Submission focus

The submitted model and notebook evaluate only the best next-hour model.